USB to EtherCAT Adapter Board V1.0

The domain of the Project:

PCB DESIGN

Team Mentors (and their designation):

Mr. C Paramesh Kumar

Hardware Design Engineer

Team Members:

Ms. Dasa Chandana

Period of the project

August 2026 to September 2026

Declaration

The project titled “USB to EtherCAT Adapter Board V1.0” has been mentored by, organised by SURE Trust, from August 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge, the members of the team mentioned below have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms, Dasa Chandana -- Signature D Chandana

Mentor’s Name
Mr. C Paramesh Kumar
Hardware Design Engineer-Wakefit Innovations

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

This project focuses on the design and development of a **USB-to-EtherCAT Adapter Board V1.0**, which is designed to establish communication between a USB host and an EtherCAT network. The board uses an STM32F411CEUx microcontroller to manage the USB interface and control functions, while the LAN9252 EtherCAT controller handles communication over the EtherCAT network. The design also includes two RJ45 connectors, an I²C EEPROM for storing non-volatile data, and a regulated 5 V to 3.3 V power supply for the different components on the board.

The development of the board started with the schematic design, where the main functional blocks and their required supporting components were identified and connected. After completing the schematic, the design was transferred to the PCB layout. Components were arranged based on their function and connection requirements, followed by routing of signal and power traces. Proper attention was given to grounding, power distribution, component placement, and signal routing to obtain a reliable PCB layout.

The complete PCB was designed using KiCad. The design was then checked for possible errors and reviewed using the 3D viewer to verify the component placement and overall board arrangement. The project provided practical experience in the complete PCB development process, including component selection, schematic design, PCB layout, routing, and design verification. It also helped in understanding the practical considerations involved in designing a communication-based board that combines USB, EtherCAT, memory, and power management circuits on a single board.

Background and Context of the Project

Modern industrial automation depends on fast and dependable communication between controllers, computers, and field devices. As automated systems become more connected, communication interfaces need to support reliable data transfer with low delay and consistent performance. **EtherCAT (Ethernet for Control Automation Technology)** is widely used in industrial applications because it is designed for high-speed and deterministic communication between networked devices.

A computer or development system may not always have a direct interface for communicating with an EtherCAT network. This creates a need for a dedicated interface that can connect the host system to EtherCAT-based devices. To address this requirement, this project focuses on the design of a **USB-to-EtherCAT Adapter Board V1.0**.

The proposed board integrates the required USB interface, STM32F411CEUx microcontroller, LAN9252 EtherCAT controller, RJ45 connectors, EEPROM, clock circuitry, and power supply on a single PCB. The project covers the development of these functional blocks and their integration into one board, starting from the schematic and component selection and continuing through PCB layout and design verification. This provides practical experience in designing a communication-based PCB for industrial applications.

Problem Statement and Goals

The main goal of this project is to design a **USB-to-EtherCAT Adapter Board V1.0** that can provide an interface between a USB host and an EtherCAT network. The board brings together the major circuits required for this purpose, including the STM32F411CEUx microcontroller, LAN9252 EtherCAT controller, USB interface, RJ45 network interfaces, EEPROM, clock circuit, and power supply.

The project aims to develop the complete PCB design and properly integrate these functional blocks on a single board. Particular attention is given to the connections between the microcontroller and EtherCAT controller, power requirements, signal routing, and the overall organization of the PCB.

Scope and Limitations

The scope of the project includes schematic development, component selection, PCB layout, component placement, signal routing, power and ground distribution, and design verification using KiCad. Supporting circuits such as the 5 V to 3.3 V voltage regulation, I²C EEPROM, clock circuitry, and dual RJ45 interfaces are also included in the design.

The project is mainly focused on the **hardware and PCB design aspects** of the adapter board. It does not cover the development of a complete commercial-grade EtherCAT software stack or extensive real-world performance and reliability testing of the manufactured board.

Practical Aspect

The practical part of the project involves bringing several different functional circuits together on a single PCB. This includes working with USB communication, microcontroller interfacing, EtherCAT communication, memory, clock generation, and power regulation.

Project Objectives

The main objective of this project is to design a **USB-to-EtherCAT Adapter Board V1.0** that integrates the required hardware for communication between a USB host and an EtherCAT network. The project aims to develop the complete PCB design while considering the electrical and physical requirements of the different functional blocks.

The specific objectives are to:

- Design the schematic integrating the STM32F411CEUx microcontroller and LAN9252 EtherCAT controller.
- Provide the required USB and dual RJ45 interfaces for host and EtherCAT connectivity.
- Design the 5 V to 3.3 V power supply, clock, reset, EEPROM, and supporting circuits.
- Develop the PCB layout with appropriate component placement, signal routing, power distribution, and grounding.
- Perform design-rule checking **(DRC)** and review the completed layout for possible design issues.
- Generate a 3D representation of the completed PCB to verify component placement and overall board arrangement.

Expected Outcomes and Deliverables

The project is expected to result in a **complete and well-organized PCB design** for the USB-to-EtherCAT Adapter Board V1.0. The final design should bring together all the required functional blocks and provide a clear basis for further fabrication and testing.

The main deliverables of the project include the circuit schematic, PCB layout, KiCad design files, design verification results, and 3D model/visualization of the completed board.

Methodology:

The PCB design was carried out in stages, starting with the requirements of the USB-to-EtherCAT Adapter Board and identification of the main circuit blocks. Components such as the STM32F411CEUx, LAN9252 EtherCAT controller, USB interface, RJ45 connectors, EEPROM, and power-supply circuit were selected based on the design requirements. Their datasheets were referred to for pin configurations, voltage levels, and recommended connections.

The circuit schematic was developed using KiCad, including supporting components such as resistors, decoupling capacitors, clock, reset, and power-filtering circuits. After checking the schematic, suitable footprints were assigned and the design was transferred to the PCB layout.

Components were arranged according to their functional sections, followed by routing of signal, power, and ground connections. The completed layout was checked using Design Rules Check (DRC), and the 3D Viewer was used to verify component placement and the overall board arrangement. Finally, the Gerber files were generated for potential PCB fabrication.

Overall, the project covered the main stages of PCB development, from component selection and schematic design to layout, routing, verification, and Gerber generation.

Tools/Software used:

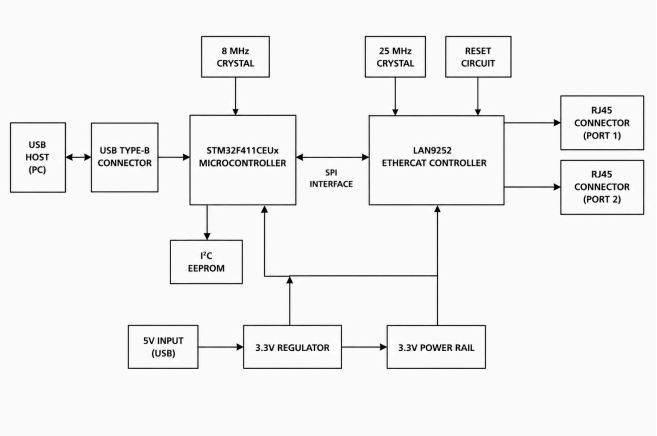


The development of the USB-to-EtherCAT adapter PCB requires tools for schematic design, component selection, PCB layout, verification, and reference of electrical specifications. The following tools and resources were used during the project:

- **KiCad:** Used as the main PCB design software for creating the schematic, assigning component footprints, designing the PCB layout, routing tracks, performing design-rule checks, and generating the 3D visualization of the completed board.
- **KiCad Schematic Editor:** Used to create the circuit diagram and establish the electrical connections between the STM32F411CEUx, LAN9252, USB interface, RJ45 interfaces, EEPROM, power supply, and supporting components.
- **KiCad PCB Editor:** Used for component placement, track routing, vias, copper zones, board outline creation, and overall PCB layout development.
- **KiCad 3D Viewer:** Used to inspect the physical appearance of the PCB and verify the placement and orientation of components before fabrication.
- **Component Datasheets:** Used to obtain important information such as pin configurations, operating voltage, electrical characteristics, recommended component values, and application requirements.
- **Manufacturer Reference Documentation:** Used to understand the recommended implementation and supporting circuitry for major components such as the **LAN9252 EtherCAT controller and STM32F411CEUx microcontroller**.

Project Architecture:

The USB-to-EtherCAT adapter is organized around two main processing and communication devices: the **STM32F411CEUx microcontroller** and the **LAN9252 EtherCAT controller**. The STM32F411 provides the interface on the USB side, while the LAN9252 handles the EtherCAT communication and connects to the external network through the RJ45 interfaces.



The major functional blocks of the system are:

- **USB Interface:** Provides the connection between the host computer and the adapter board and supplies the board with 5 V power.
- **STM32F411CEUx Microcontroller:** Acts as the main microcontroller on the USB side and manages the interface and control functions of the board.
- **LAN9252 EtherCAT Controller:** Handles the EtherCAT communication and provides the interface between the microcontroller and the EtherCAT network.
- **EtherCAT / RJ45 Interface:** Two RJ45 connectors provide the external network connections required for EtherCAT communication.
- **I²C EEPROM:** Provides non-volatile memory for storing required configuration or device-related information.
- **Power Supply:** The incoming 5 V supply is regulated to **3.3 V** to power the required digital circuits on the board.

- **Clock and Supporting Circuits:** Crystal circuits, reset circuitry, decoupling capacitors, resistors, and other supporting components are provided to ensure proper operation of the major ICs.

Chapter III-Design Workflow

Schematic: \

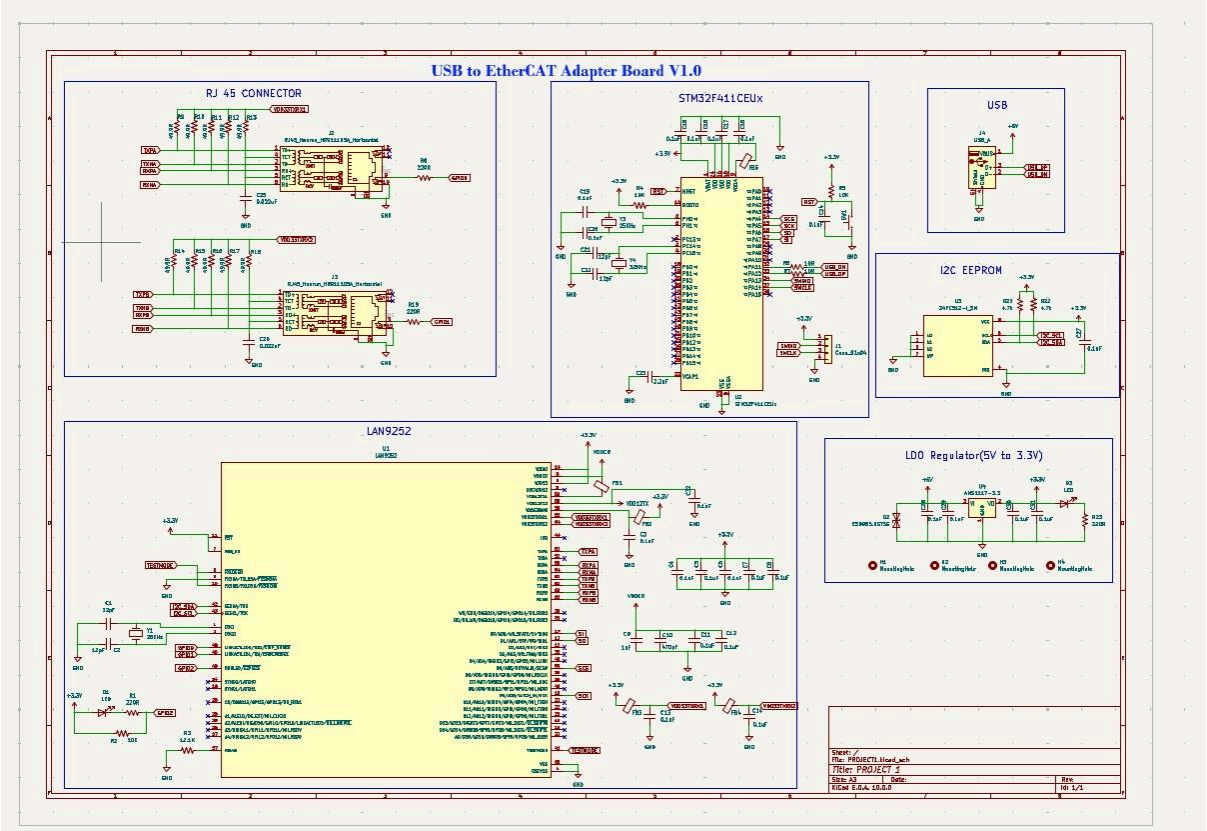


Figure (a)

Figure (a) presents the complete schematic of the USB-to-EtherCAT Adapter Board V1.0. The circuit is organized into different functional sections, including the STM32F411CEUx microcontroller, LAN9252 EtherCAT controller, USB interface, RJ45 interfaces, and supporting circuits. These sections are interconnected to provide the required communication and control functions of the adapter board.

5.2 USB Interface

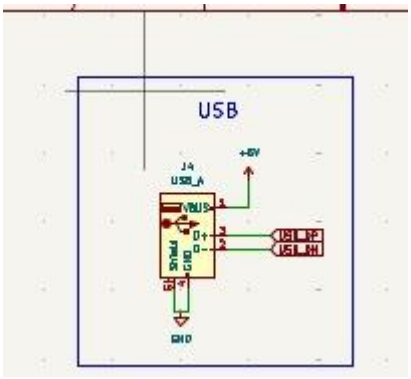


Figure (b)

The USB Type-A connector provides the connection between the host computer and the adapter board. It carries USB data to the STM32F411CEUx microcontroller and also supplies the 5 V input required by the board.

The D+ and D− lines are connected to the corresponding USB pins of the microcontroller for data transfer. The VBUS (5 V) line feeds the board's power supply section, while the USB ground provides the common ground reference for the circuit.

LAN9252 EtherCAT Controller:

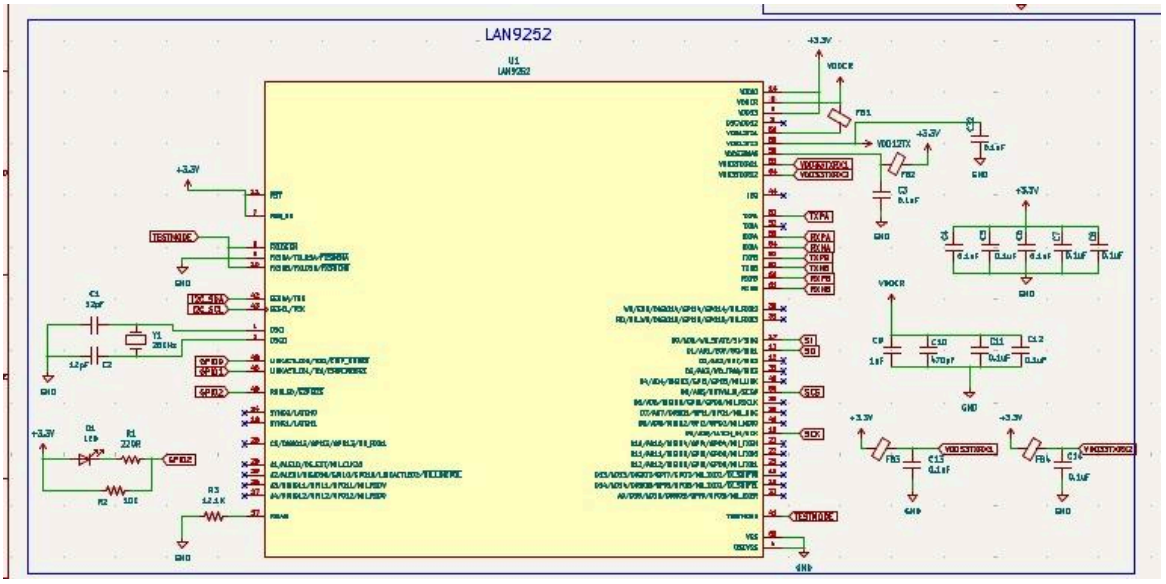


Figure (c)

The **LAN9252** is the main EtherCAT controller used on the PCB and acts as the communication link between the STM32F411CEUx microcontroller and the EtherCAT network. It manages EtherCAT communication and provides the required interfaces for connecting the board to the external network through the two RJ45 ports. The LAN9252 section also includes the supporting clock, reset, power, decoupling, and interface circuitry required for proper operation.

Key Specifications:

- **Device:** LAN9252 EtherCAT Controller
- **Function:** EtherCAT Slave Controller
- **Supply Voltage:** 3.3 V
- **Host Interface:** SPI / Digital Host Interface with STM32
- **Ethernet Ports:** Two — EtherCAT IN and OUT
- **Interface:** EtherCAT over Ethernet physical layer
- **Clock:** External crystal/clock circuitry as required
- **Supporting Circuitry:** Decoupling capacitors, reset circuitry, clock circuit, and required passive components

STM32F411CEUx Microcontroller

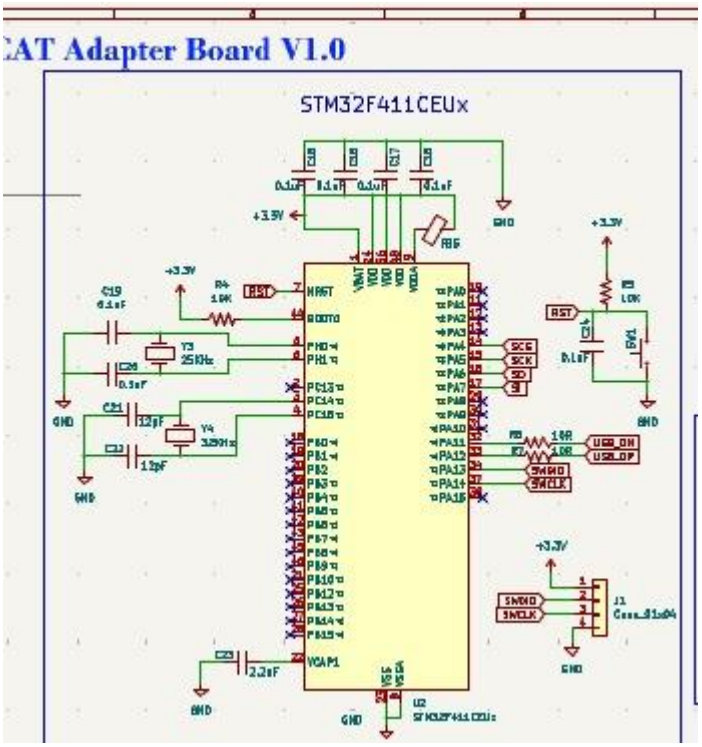


Figure (d)

The **STM32F411CEUx** is the main microcontroller on the USB side of the adapter. It handles communication with the USB host and interfaces with the LAN9252 EtherCAT controller through the required digital connection.

The schematic includes the supporting circuits needed for reliable operation, such as power decoupling, clock generation, reset and boot configuration, USB connections, and an SWD interface for programming and debugging.

Key Specifications:

- **Microcontroller:** STM32F411CEUx
- **Maximum operating frequency:** Up to 100 MHz
- **Core:** ARM Cortex-M4 with Floating Point Unit (FPU)
- **SRAM:** 128 KB
- **Flash memory:** 512 KB

5 V to 3.3 V LDO Regulator

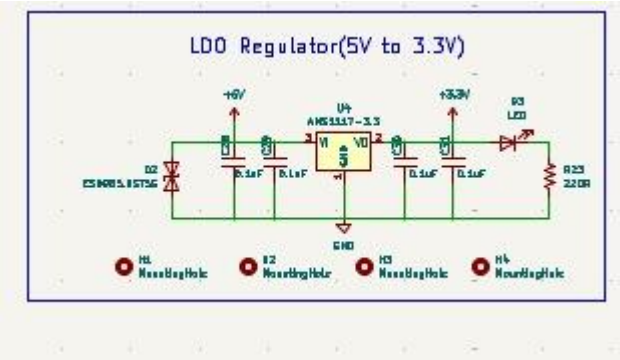


Figure (e)

The **5 V to 3.3 V power supply section** provides the regulated 3.3 V supply required by the board's digital components. The AMS1117-3.3 LDO regulator converts the 5 V input into a stable 3.3 V output.

Input and output capacitors are used for filtering and regulator stability, while an ESD protection device is provided at the input for additional protection. A power LED is also included to indicate the presence of the 3.3 V supply.

Key Specifications:

- **Regulator:** AMS1117-3.3
- **Input protection:** ESD protection device (D2)
- **Power indication:** LED (D3)
- **LED current-limiting resistor:** 220 Ω
- **Input voltage:** 5 V
- **Output voltage:** 3.3 V
- **Input capacitors:** 0.1 µF × 2
- **Output capacitors:** 0.1 µF × 2

I²C EEPROM

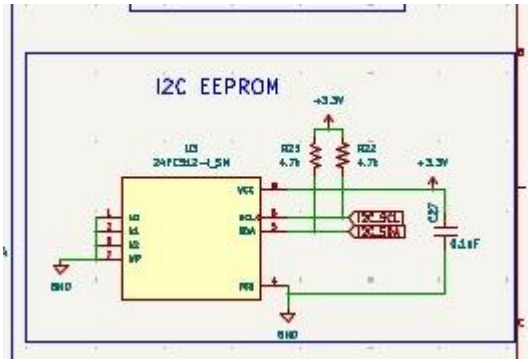


Figure (f)

The **I²C EEPROM** is used to store configuration data and other information that needs to be retained even when the board is powered off. The design uses the 24FC512-I/SM, which communicates with the system through the I²C interface.

The SCL and SDA lines are connected to the system with suitable pull-up resistors. The EEPROM address pins and write-protect (WP) pin are connected to ground to set the required device configuration and allow normal memory operation.

Key Specifications:

- **EEPROM:** 24FC512-I/SM
- **Supply voltage:** 3.3 V
- **I²C lines:** SCL and SDA
- **Pull-up resistors:** 4.7 kΩ for SCL and SDA
- **Address pins (A0, A1, A2):** Connected to GND
- **Memory capacity:** 512 Kbit
- **Interface:** I²C

RJ45 / EtherCAT Interface

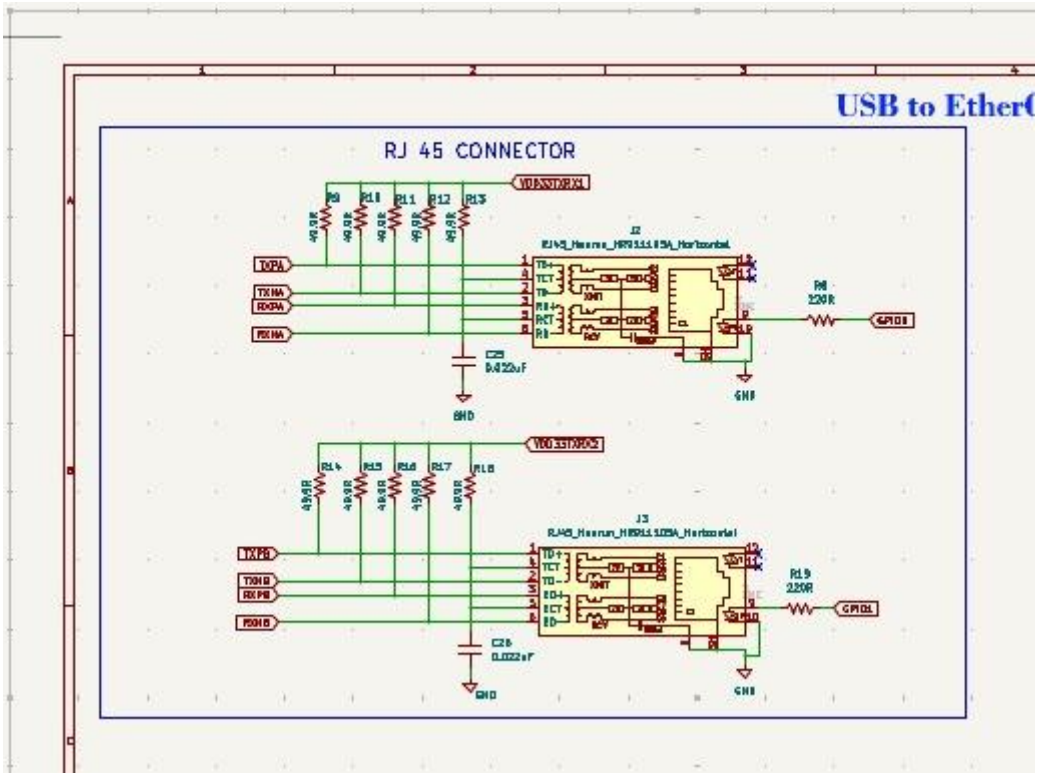


Figure (g)

The **RJ45 interface section** provides the physical connection between the LAN9252 EtherCAT controller and the external EtherCAT network. The design uses two RJ45 connectors, corresponding to the two EtherCAT ports, with each port carrying separate transmit and receive differential signals. The connectors include integrated Ethernet magnetics and status LED connections, while the surrounding termination and filtering components help provide the required electrical interface for reliable Ethernet communication.

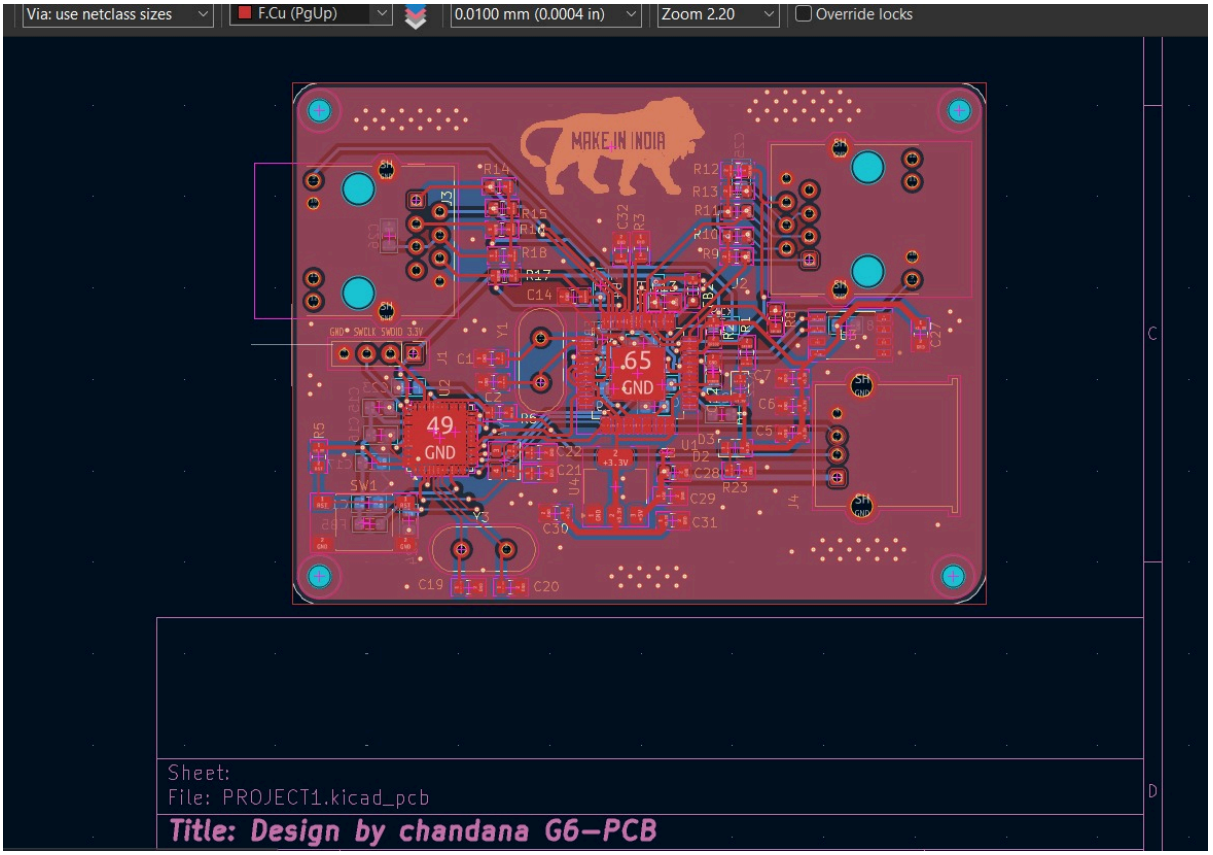
Key Specifications:

- **Number of RJ45 ports:** 2
- **Interface:** 10/100 Mbps Ethernet / EtherCAT
- **Termination resistors:** 49.9 Ω resistors used on the differential signal paths

- **Filtering:** 0.022 µF capacitors provided in the interface section
- **Supply:** 3.3 V Ethernet interface supply
- **Status indication:** LED connections provided through GPIO signals
- **LED resistors:** 220 Ω current-limiting resistors
- **Ethernet signals:** TXP, TXN, RXP and RXN for each port
- **Magnetics:** Integrated in the RJ45 connector

The board includes **two RJ45 ports** that provide the **EtherCAT IN and OUT connections**, allowing the adapter to be connected within an EtherCAT network. The interfaces include the required Ethernet magnetics and termination components for the physical-layer connection. These components help provide proper signal coupling and maintain reliable signal integrity during high-speed differential communication.

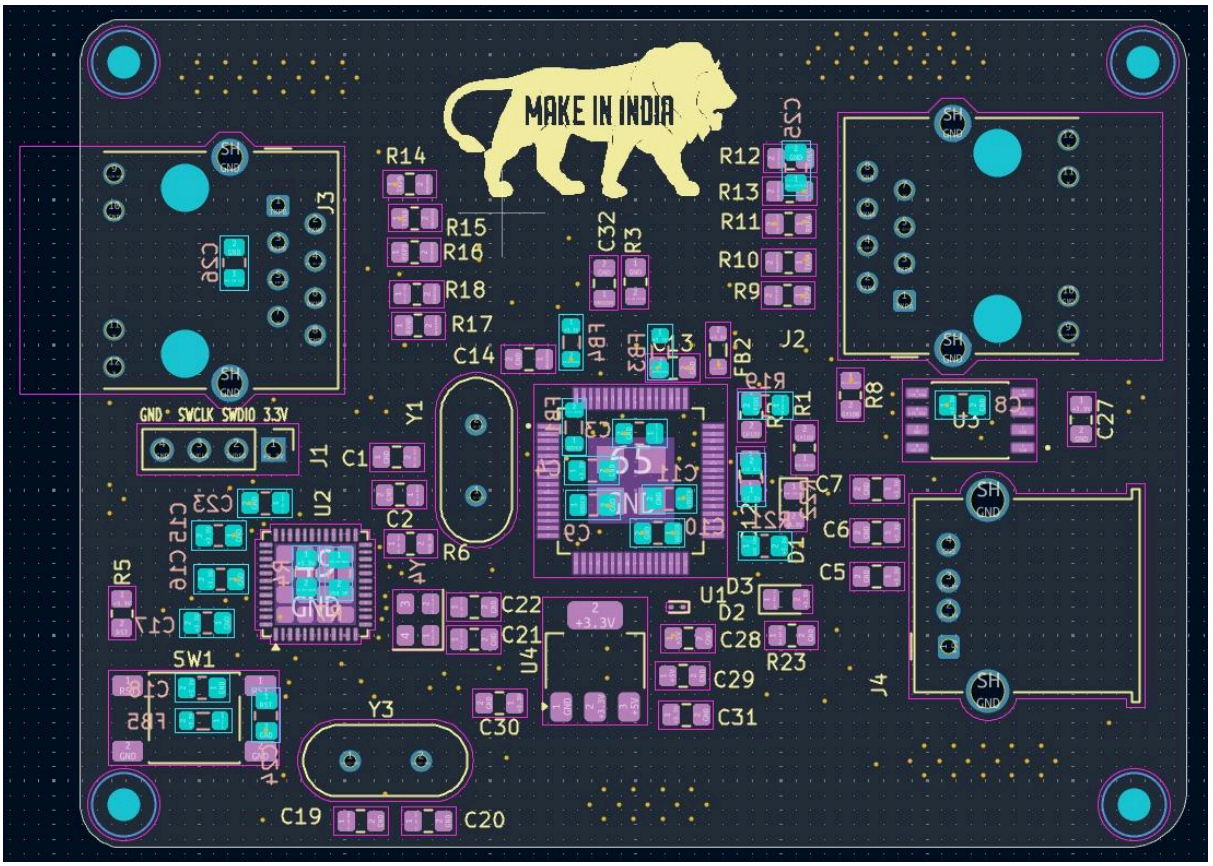
PCB Layout



The completed PCB layout of the **USB-to-EtherCAT Adapter Board V1.0**, developed using KiCad. The layout includes the major components and supporting circuits arranged according to their functional sections. The board contains the microcontroller, EtherCAT controller, power supply, USB interface, EEPROM, and RJ45 interfaces.

The PCB layout includes signal routing, power and ground connections, mounting holes, and copper areas. Component placement and routing were carried out to keep the design organized and provide proper connections between the different sections of the board. The

Component Placement



Placement Considerations:

- **LAN9252 – Center:**

  The **LAN9252 EtherCAT controller** is placed near the center of the board as the main communication device. This position helps keep its connections to the network interfaces and the STM32 relatively organized.

- **STM32F411CEUx – Left-center:**

  The **STM32F411CEUx** is positioned on the left-center portion of the board. Its supporting components, including the crystal, decoupling capacitors, and other passive components, are placed nearby to keep the connections short and organized.

- **RJ45 Interfaces – Board edges:**

  The two RJ45 interfaces are positioned near the **left and right edges** of the PCB. Placing the connectors along the board edges makes them easily accessible for external EtherCAT connections.

- **EtherCAT supporting components – Near LAN9252 and RJ45 interfaces:**
  The resistors, capacitors, and other supporting components associated with the EtherCAT interfaces are placed close to the relevant circuits. This helps reduce unnecessary trace lengths and keeps the high-speed communication paths organized.
- **Crystals – Close to respective ICs:**
  The crystal components are placed close to the ICs that require clock signals. Keeping these connections short helps reduce unwanted noise and interference in the clock signals.
- **Decoupling capacitors – Close to ICs:**
  Decoupling capacitors are placed close to the power pins of the **STM32 and LAN9252** to provide local power filtering and help reduce supply noise.
- **LDO Regulator – Lower-center:**
  The **AMS1117-3.3** power supply section is positioned in the lower-center area of the board. Keeping the power components grouped together makes the power section easier to organize and route.
- **SWD Connector – Left side:**
  The **SWD programming/debug connector** is placed on the left side of the PCB, providing convenient access for programming and debugging the STM32.
- **Mounting Holes – Four corners:**
  Four mounting holes are provided near the corners of the PCB to provide mechanical support and allow the board to be securely mounted.

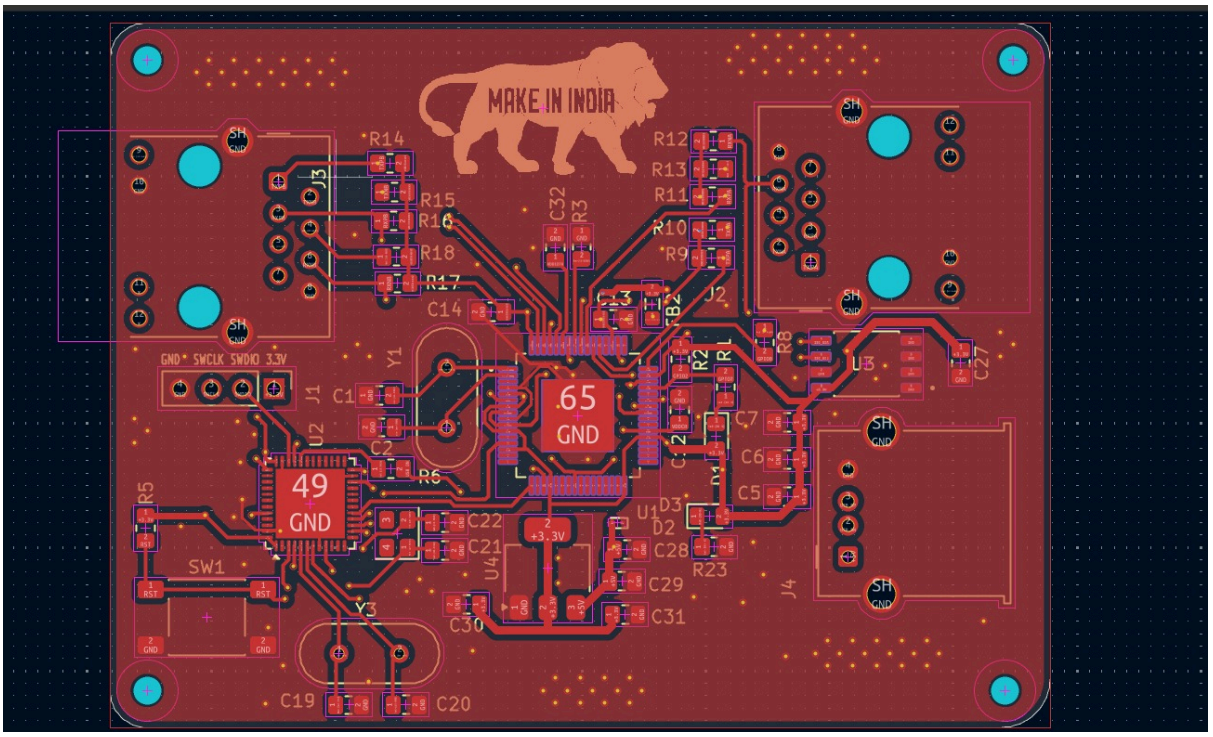


Figure (h): Front Copper layer

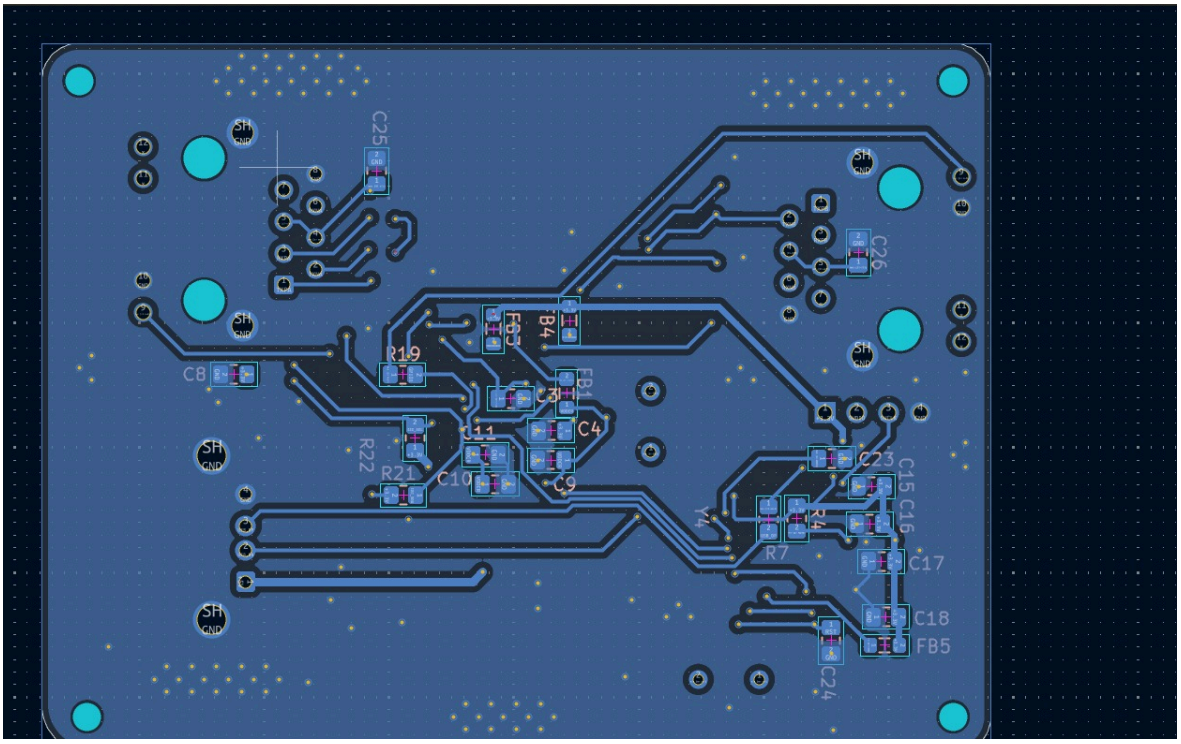
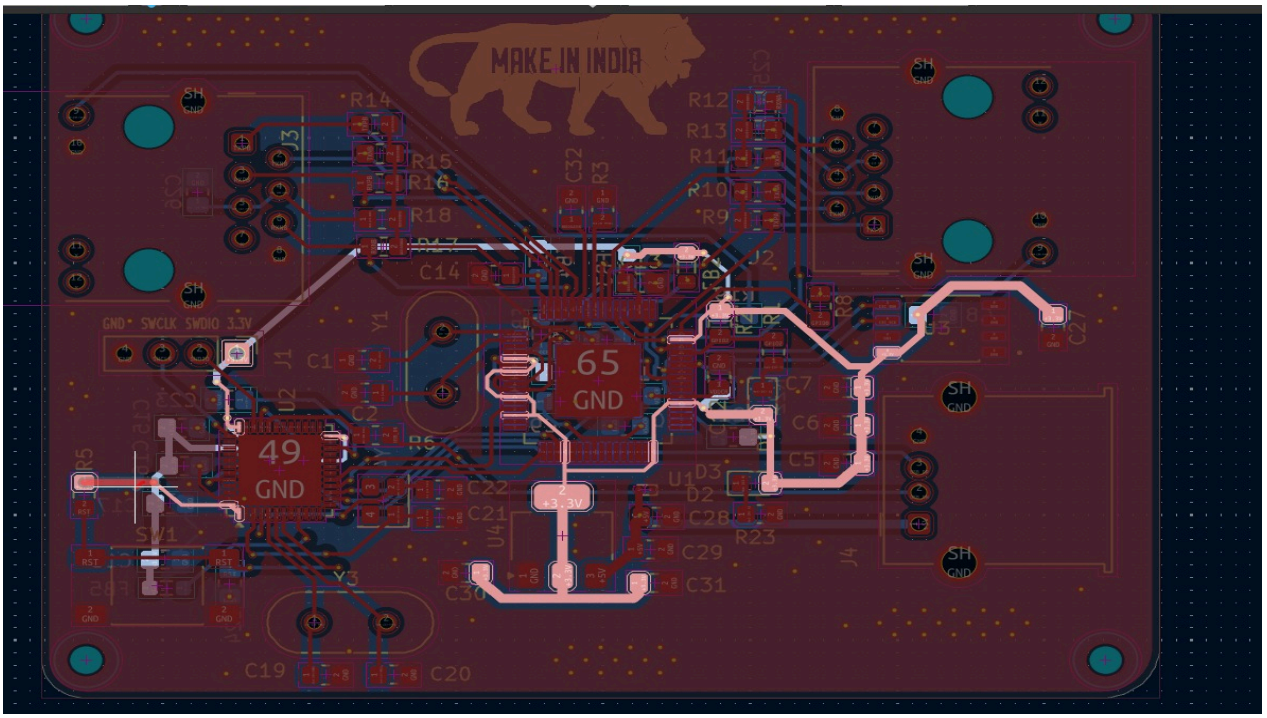
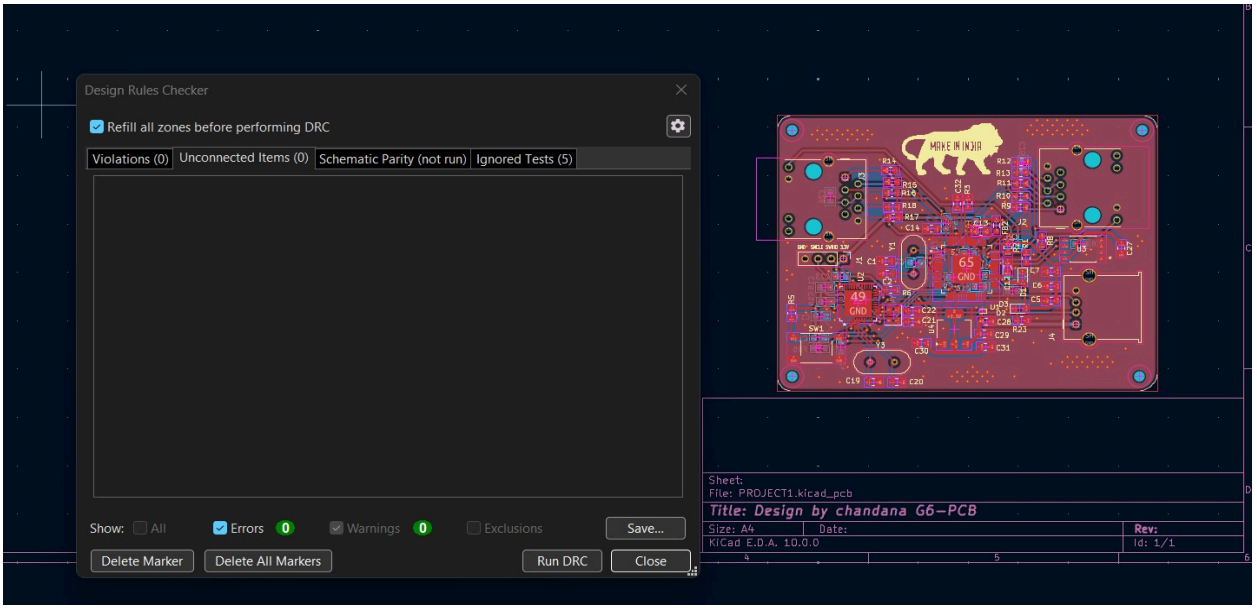


Figure (i): Back Copper layer



3.3V Conducting Path



Verified Design using Design Rules Checker(DRC)

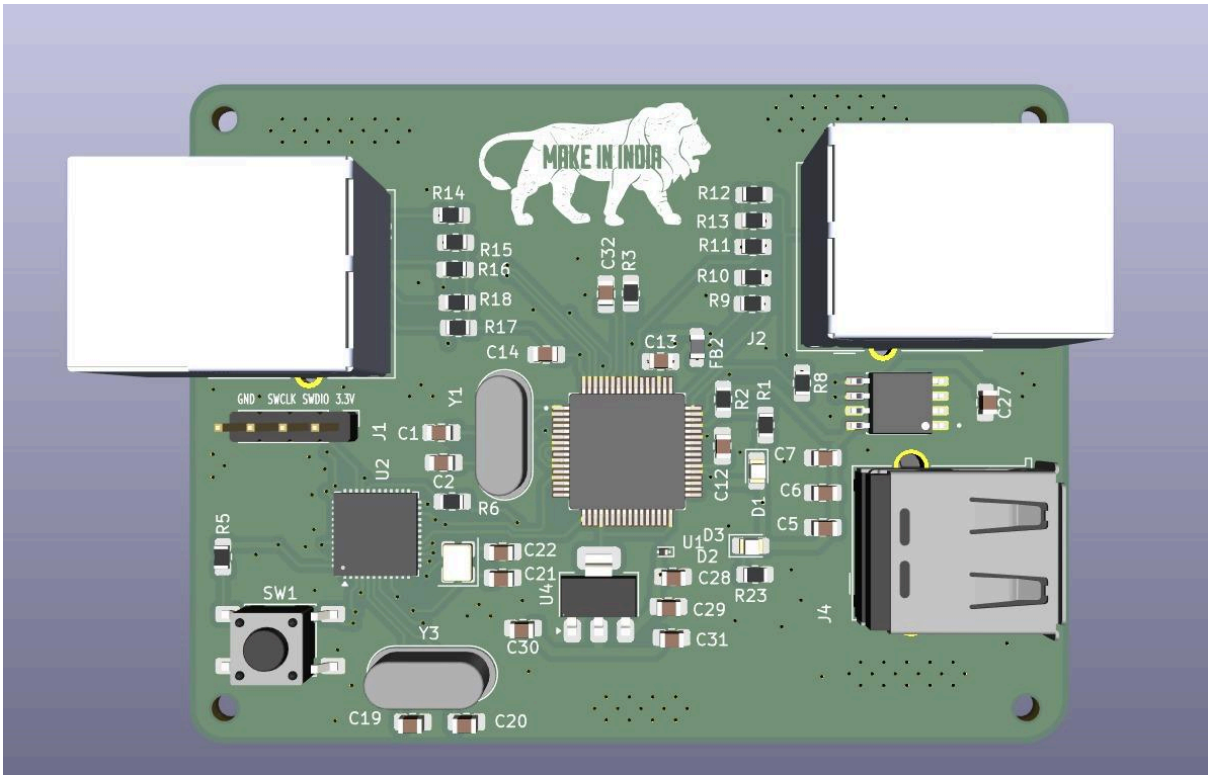


Fig. Front 3D view

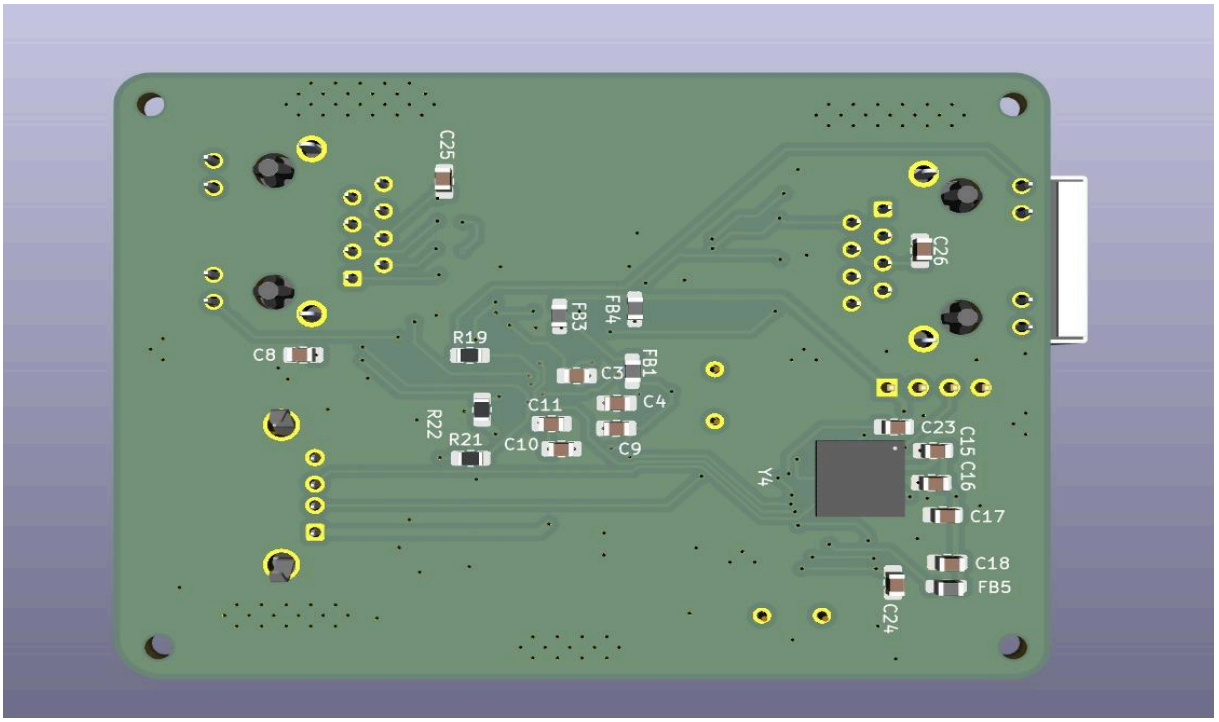


Fig. Back 3D view

Social/Industry Relevance

The **USB-to-EtherCAT Adapter Board** is useful in industrial automation, where different machines and devices need to communicate with each other quickly and reliably. EtherCAT is commonly used in areas such as robotics, CNC machines, motion control, and factory automation.

This project provides a way to connect a computer or embedded system through USB to an EtherCAT network. This can be useful for testing, monitoring, configuring, and communicating with EtherCAT-based devices without requiring a computer to have a built-in EtherCAT interface.

The project is also relevant from a learning and industry point of view because it involves **PCB design, USB communication, Ethernet communication, hardware interfacing, and embedded systems**. These are commonly used skills in industrial automation and electronics companies. Overall, the project gives practical experience in designing a hardware interface that can be used in real-world industrial communication systems.

This project provided practical experience in developing a PCB from the initial **schematic to the final layout**. Working on the board helped me understand how different sections, including the **STM32F411CEUx, LAN9252, USB, RJ45 interfaces, EEPROM, and power supply**, are integrated into a single design. It also showed the importance of proper component placement, routing, grounding, and power distribution in PCB development.

Key Learnings:

- Gained hands-on experience with KiCad schematic and PCB layout.
- Learned to refer to component datasheets while designing circuits.
- Understood how to plan component placement and routing.
- Gained practical knowledge of PCB routing, vias, grounding, and power distribution.
- Learned the importance of placing decoupling capacitors and clock components close to the respective ICs.
- Understood how USB, SPI, I²C, and Ethernet/EtherCAT interfaces are integrated into a single PCB.
- Gained experience with ERC, DRC, and 3D PCB visualization in KiCad..

Reflections:

One of the main lessons from this project was that PCB design requires both a good understanding of the circuit and careful planning of the physical layout. Even small decisions, such as component placement and trace routing, can affect the overall arrangement and reliability of the board. Working through the complete design process also helped me understand how the concepts learned in electronics are applied in a practical PCB development environment.

Conclusion

The project involved the complete design of a USB-to-EtherCAT Adapter Board V1.0, starting from the schematic and progressing through component selection, PCB placement, routing, and design verification. The final design brings together the STM32F411CEUx, LAN9252 EtherCAT controller, USB interface, dual RJ45 interfaces, I²C EEPROM, and 5 V to 3.3 V power supply on a single PCB. Working through these stages provided practical experience in PCB development and helped in understanding the design considerations involved in building a communication-oriented board.

Future Scope

The designed PCB can be further developed from a design-stage prototype into a **fabricated and tested hardware board**. Future work can include manufacturing the PCB, assembling the components, and performing electrical and functional testing of the USB and EtherCAT interfaces.

The following improvements can be considered for further development of the project:

- Fabricating the PCB and performing hardware testing to verify the design.
- Testing USB-to-EtherCAT communication with a host computer and EtherCAT devices.
- Performing signal integrity and power-supply measurements on the fabricated board.
- Making necessary PCB layout improvements based on the test results.
- Developing the required firmware and software to achieve complete USB-to-EtherCAT communication.

Project Files

The complete project files, including the schematic, PCB layout, are provided in the following Google Drive folder.

Google Drive: *[Project Files – USB-to-EtherCAT Adapter Board V1.0]*

Previous PCB Designs

S.No	Title of Project	Design Files Link
1	LED Design	Click here
2	LDR Design	Click here
3	DC to DC Converter	Click here
4	USB to TTlL Converter	Click here
5	TSH82 Op Amp	Click here
6	8051 Microcontroller	Click here
7	Arduino Pro Mini	Click here
8	Arduino 2.0	Click here
9	Arduino 3.0	Click here
10	STM32 board	Click here